



**Muhammad
Mehtab**

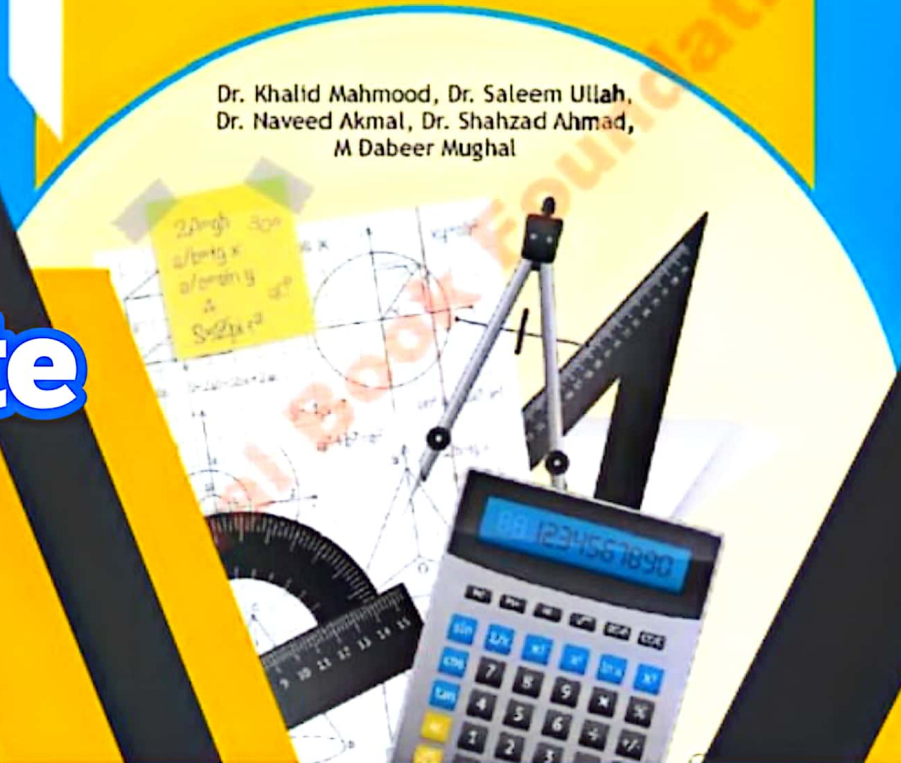
1.1

Complete

Exercise
Complex Numbers

11
Model Textbook of
MATHEMATICS

Dr. Khalid Mahmood, Dr. Saleem Ullah,
Dr. Naveed Akmal, Dr. Shahzad Ahmad,
M Dabeer Mughal

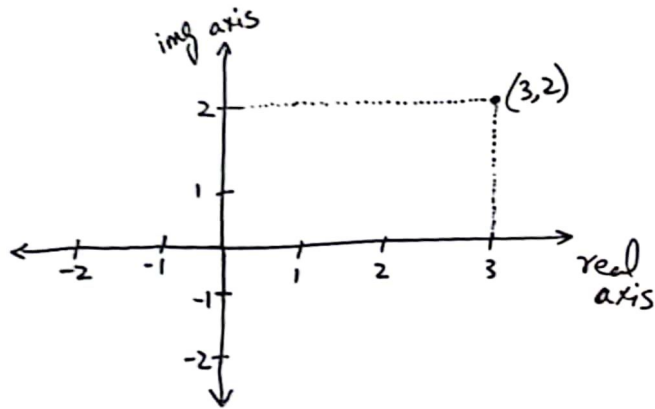


UNIT #1 EXERCISE #1.1

Standard Form

$$Z = x + yi \quad , x, y \in \mathbb{R}$$

Example: $Z = 3 + 2i$



Graph: Complex Number

Theorem: Every Real Number is a Complex Number.

Example $Z = 5 + 0i$

Converse Not True.

COMPLEX NUMBER

ordered pair form

$$Z = (x, y)$$

$$i = ?$$

let Consider Standard Quadratic Equation

$$ax^2 + bx + c = 0$$

Put $a=1, b=0, c=1$

$$x^2 + 1 = 0 \quad \text{Pure Quadratic Equation.}$$

Now Solve this Equation

$$\sqrt{x^2} = \sqrt{-1}$$

$$x = \sqrt{-1} = i \quad \text{is called (iota)}$$

$$\Rightarrow \boxed{i^2 = -1}$$

UNIT #1 EXERCISE #1.1

(12) Evaluate the following.

$$\begin{aligned} \text{(i)} \quad & i^{31} \\ &= (i^2)^{15} \cdot i \\ &= (-1)^{15} \cdot i \\ &= -i \end{aligned}$$

Answer

$$\begin{aligned} \text{(ii)} \quad & (-i)^6 \\ &= + i^6 \\ &= (i^2)^3 \\ &= (-1)^3 \\ &= -1 \end{aligned}$$

Answer.

$$\begin{aligned} \text{(iii)} \quad & (-1)^{\frac{-13}{2}} \\ &= \left[(-1)^{\frac{1}{2}} \right]^{-13} \\ &= (\sqrt{-1})^{-13} \\ &= (i)^{-13} \\ &= \frac{1}{i^{13}} \\ &= \frac{1}{(i^2)^6 \cdot i} \end{aligned}$$

COMPLEX NUMBER

$\therefore i^2 = -1$

$(-1)^{\text{odd}} = -$
 $(-1)^{\text{even}} = +$

$$\begin{aligned} &= \frac{1}{(-1)^6 \cdot i} \\ &= \frac{1}{i} \times \frac{-i}{-i} \\ &= \frac{-i}{-i^2} \\ &= \frac{-i}{-(-1)} \\ &= \frac{-i}{1} = -i \end{aligned}$$

Answer.

UNIT #1 EXERCISE #1.1

COMPLEX NUMBER

① Evaluate the following:

$$(iv) \frac{2}{(-1)^{3/2}}$$

$$\begin{aligned} \text{Sol: } &= \frac{2}{[(-1)^{1/2}]^3} \\ &= \frac{2}{[\sqrt{-1}]^3} \\ &= \frac{2}{i^3} \\ &= \frac{2}{i^2 \cdot i} \end{aligned} \quad \begin{aligned} &= \frac{2}{-i} \times \frac{+i}{+i} \\ &= \frac{2i}{-i^2} \\ &= \frac{2i}{-(-1)} \\ &= \frac{2i}{1} \\ &= 2i \end{aligned}$$

Answer

$$\begin{aligned} (v) \quad &i^{23} + i^{58} + i^{21} \\ &= (i^2)^{11} \cdot i + (i^2)^{29} + (i^2)^{10} \cdot i \\ &= (-1)^{11} \cdot i + (-1)^{29} + (-1)^{10} \cdot i \\ &= -i - 1 + i \\ &= -1 \end{aligned}$$

Answer

$$\because i^2 = -1$$

$$\sqrt{i^2} = \sqrt{-1}$$

$$\therefore i = \sqrt{-1}$$

$$(-1)^{\text{odd}} = -1$$

$$(-1)^{\text{even}} = +1$$

UNIT #1 EXERCISE #1.1

COMPLEX NUMBER

Q2 Write the following Complex Numbers in the form $x+yi$.

$$\because i^2 = -1$$

$$\begin{aligned} (-1)^{\text{odd}} &= - \\ (-1)^{\text{even}} &= + \end{aligned}$$

$$\text{ii) } (3+2i) + (2+4i)$$
$$= 5+6i$$

Answer

$$\text{iii) } (4+7i) + (4-7i)$$
$$= \frac{8+0i}{8}$$

Answer

$$\text{iv) } (3+2i)(4-3i)$$

$$= 3(4-3i) + 2i(4-3i)$$

$$= 12 - 9i + 8i - 6i^2$$

$$= 12 - i - 6(-1)$$

$$= 12 - i + 6$$

$$= 18 - i$$

Answer

$$\text{v) } (4+3i) - (2+5i)$$

$$= 4+3i-2-5i$$

$$= 2-2i$$

Answer

$$\text{vi) } (2+5i) - (2-5i)$$

$$= 2+5i-2+5i$$

$$= \frac{0+10i}{10i}$$

$$= 10i$$

UNIT #1 EXERCISE #1.1

COMPLEX NUMBER

2) Write the following Complex Numbers in the form $x+yi$.

vi) $(3, 2) \div (3, -1)$

$$\begin{aligned}\text{Sol} &= \frac{3+2i}{3-i} \times \frac{3+i}{3+i} \\ &= \frac{3(3+i) + 2i(3+i)}{(3)^2 - i^2} \\ &= \frac{9+3i+6i+2i^2}{9-(-1)} \\ &= \frac{9+9i-2}{9+1}\end{aligned}$$

$$= \frac{7+9i}{10}$$

$$= \frac{7}{10} + \frac{9}{10}i$$

Answer

vii) $(1+i)(1-i)(2+i)$

$$= (1-i^2)(2+i)$$

$$= (1+1)(2+i)$$

$$= 2(2+i)$$

$$= 4+2i$$

Answer

viii) $\frac{1}{2+3i}$

$$= \frac{1}{2+3i} \times \frac{2-3i}{2-3i}$$

$$= \frac{2-3i}{4-9i^2}$$

$$= \frac{2-3i}{4+9}$$

$$= \frac{2}{13} - \frac{3}{13}i$$

Answer

$$\because i^2 = -1$$

$$(-1)^{\text{odd}} = -1$$

$$(-1)^{\text{even}} = +1$$

UNIT #1 EXERCISE #1.1

2 Simplify the following

$$1. \frac{(2+i)(3-2i)}{(1+i)}$$

sol.

$$\frac{2(3-2i) + i(3-2i)}{1+i}$$

$$\frac{6-4i+3i-2i^2}{1+i}$$

$$\frac{-i-2(-1)}{1+i}$$

$$\frac{i+2}{1+i}$$

$$\frac{8-i}{1+i}$$

$$= \frac{8-i}{1+i} \times \frac{1-i}{1-i}$$

$$= \frac{8(1-i) - i(1-i)}{1-i^2}$$

$$= \frac{8-8i-i+i^2}{1-(-1)}$$

$$= \frac{8-9i+(-1)}{1+1}$$

$$= \frac{7-9i}{2}$$

$$= \frac{7}{2} - \frac{9}{2}i$$

Answer

$$(a+b)^2 = a^2 + b^2 + 2ab$$

COMPLEX NUMBER

$$\because i^2 = -1$$

$$(-1)^{\text{odd}} = -$$

$$(-1)^{\text{even}} = +$$

$$(ii) \frac{(1+i)}{(2+i)^2}$$

$$= \frac{1+i}{4+i^2+4i}$$

$$= \frac{1+i}{4-1+4i}$$

$$= \frac{1+i}{3+4i} \times \frac{3-4i}{3-4i}$$

$$= \frac{1(3-4i) + i(3-4i)}{9-16i^2}$$

$$= \frac{3-4i+3i-4i^2}{9-16(-1)}$$

$$= \frac{3-i-4(-1)}{9+16}$$

$$= \frac{3-i+4}{25}$$

$$= \frac{7}{25} - \frac{1}{25}i$$

Answer

UNIT #1 EXERCISE #1.1

③ Simplify the following

$$\text{iii) } \frac{1}{3+i} - \frac{1}{3-i}$$

$$= \frac{1}{3+i} \times \left(\frac{3-i}{3-i} \right) - \frac{1}{3-i} \times \left(\frac{3+i}{3+i} \right)$$

$$= \frac{3-i}{9-i^2} - \frac{3+i}{9-i^2}$$

$$= \frac{3-i}{9+1} - \frac{3+i}{9+1}$$

$$= \frac{3-i}{10} - \frac{3+i}{10}$$

$$= \frac{\cancel{3}-i-\cancel{3}-i}{10}$$

$$= \frac{-2i}{10}$$

$$= 0 - \frac{1}{5}i$$

Answer

COMPLEX NUMBER

$$\therefore \boxed{i^2 = -1}$$
$$(-1)^{\text{odd}} = -$$
$$(-1)^{\text{even}} = +$$

$$(iv) (1+i)^{-2} + (1-i)^{-2}$$

$$= \frac{1}{(1+i)^2} + \frac{1}{(1-i)^2}$$

$$= \frac{1}{1+i^2+2i} + \frac{1}{1+i^2-2i}$$

$$= \frac{1}{1-1+2i} + \frac{1}{1-1-2i}$$

$$= \frac{1}{\cancel{2}i} - \frac{1}{\cancel{2}i}$$

$$= 0 + 0i$$

Answer

UNIT #1 EXERCISE #1.1

③ Simplify the following

$$v) \frac{(2+i)^2 + (7-4i)}{(2+i)}$$

$$\begin{aligned} &= (4 + i^2 + 4i) + \frac{7-4i}{2+i} \times \frac{2-i}{2-i} \\ &= (4-1+4i) + \frac{7(2-i) - 4i(2-i)}{4-i^2} \\ &= (3+4i) + \frac{14-7i-8i+4i^2}{4+1} \\ &= (3+4i) + \frac{14-15i-4}{5} \\ &= (3+4i) + \frac{10-15i}{5} \end{aligned}$$

COMPLEX NUMBER

$$\begin{aligned} &\because i^2 = -1 \\ &(-1)^{\text{odd}} = - \\ &(-1)^{\text{even}} = + \end{aligned}$$

$$= (3+4i) + \left(\frac{10}{5} - \frac{15}{5}i \right)$$

$$= 3+4i + 2-3i$$

$$= 5+i$$

Answer

UNIT #1 EXERCISE #1.1

COMPLEX NUMBER

(4) Find the values of the real numbers x and y in each of the following:

$$\because i^2 = -1$$

i) $(2+3i)x + (1+3i)y + 2 = 0$

Sol: $2x + 3ix + y + 3yi + 2 = 0$

$$(2x + y + 2) + (3x + 3y)i = 0 + 0i$$

$$2x + y + 2 = 0, \quad 3x + 3y = 0$$

$$2x + y = -2, \quad 3(x + y) = 0$$

$$2x + y = -2, \quad x + y = 0$$

$$2(-y) + y = -2$$

$$-y = -2$$

$$y = 2$$

Answer

ii) $\frac{x}{1+i} + \frac{y}{1-2i} = 1$

UNIT #1 EXERCISE #1.1

COMPLEX NUMBER

4) Find the values of the real numbers x and y in each of the following:

$$\because i^2 = -1$$

$$5x + 2y = 10 \quad \text{--- (1)} \quad , \quad 2y - 5x = 0 \quad \text{--- (2)}$$

Solve (1) & (2)

$$\begin{array}{r} 5x + 2y = 10 \\ + \quad -5x + 2y = 0 \\ \hline 4y = 10 \end{array}$$

$$y = \frac{10}{4} = \frac{5}{2}$$

$$\boxed{y = \frac{5}{2}} \text{ Put (2)}$$

$$2\left(\frac{5}{2}\right) - 5x = 0$$

$$5 = 5x$$

$$\frac{5}{5} = x$$

$$\Rightarrow \boxed{x = 1}$$

$$\text{(ii)} \quad \frac{x}{1+i} + \frac{y}{1-2i} = 1$$

$$\underline{\text{Sol:}} \quad \left(\frac{x}{1+i} \times \frac{1-i}{1-i} \right) + \left(\frac{y}{1-2i} \times \frac{1+2i}{1+2i} \right) = 1$$

$$\frac{x-ix}{1-i^2} + \frac{y+2yi}{1-4i^2} = 1$$

$$\frac{x-xi}{2} + \frac{y+2yi}{5} = 1$$

$$5 \left(\frac{x-xi}{2} \right) + 10 \left(\frac{y+2yi}{5} \right) = 10$$

$$5x - 5xi + 2y + 4yi = 10$$

$$(5x+2y) + (2y-5x)i = 10 + 0i \quad (4y-5x)i$$

UNIT #1 EXERCISE #1.1

COMPLEX NUMBER

④ Find the values of the scalar numbers x and y in each of the following:

$$\because i^2 = -1$$

$$\text{iii) } \frac{x}{2+i} = \frac{1-5i}{3-2i} + \frac{y}{2-i}$$

$$\text{Sol: } \frac{x}{2+i} - \frac{y}{2-i} = \frac{1-5i}{3-2i}$$

$$\frac{x(2-i) - y(2+i)}{(2+i)(2-i)} = \frac{1-5i}{3-2i} \times \frac{3+2i}{3+2i}$$

$$\frac{2x - xi - 2y - yi}{4 - i^2} = \frac{1(3+2i) - 5i(3+2i)}{9 - 4i^2}$$

$$\frac{(2x-2y) - (x+y)i}{5} = \frac{3+2i-15i-10i^2}{9+4}$$

$$\frac{(2x-2y) - (x+y)i}{5} = \frac{3-13i+10}{13}$$

$$\frac{(2x-2y) - (x+y)i}{5} = \frac{13-13i}{13}$$

$$\frac{(2x-2y) - (x+y)i}{5} = 1 - i$$

$$(2x-2y) - (x+y)i = 5 - 5i$$

Compare real & imag

$$2x-2y=5 \text{ --- (1) } \quad x+y=5 \text{ --- (3)}$$

Solve ① & ②

$$\begin{array}{r} 2x-2y=5 \\ + \quad 2x+2y=10 \\ \hline 4x=15 \end{array}$$

$$x = \frac{15}{4}$$

Put in eq ③

$$\frac{15}{4} + y = 5$$

$$y = 5 - \frac{15}{4}$$

$$= \frac{20-15}{4}$$

$$y = \frac{5}{4} \checkmark$$

Answer

NIT #1 EXERCISE #1.1

COMPLEX NUMBER

4) Find the values of the real numbers x and y in each of the following:

$$\because i^2 = -1$$

$$\text{iv) } x(1+i)^2 + y(2-i)^2 = 3+10i$$

$$\text{sol. } x(1+i^2+2i) + y(4+i^2-4i) = 3+10i$$

$$x(1+1+2i) + y(4-1-4i) = 3+10i$$

$$2xi + y(3-4i) = 3+10i$$

$$2xi + 3y - 4yi = 3+10i$$

$$3y + (2x-4y)i = 3+10i$$

Compare real & imag

$$3y = 3 \quad , \quad 2x - 4y = 10 \quad \text{--- (1)}$$

$$\boxed{y=1} \quad \checkmark \quad \text{Put in (1)}$$

$$2x - 4(1) = 10$$

$$2x - 4 = 10$$

$$2x = 10 + 4$$

$$2x = 14$$

$$x = \frac{14}{2}$$

$$\boxed{x=7}$$

✓
Answer

UNIT #1 EXERCISE #1.1

COMPLEX NUMBER

⑤ Find the Complex Number Z if

$$4Z - 3\bar{Z} = \frac{1-18i}{2-i}$$

Sol: let $Z = x + yi$

$$4(x+yi) - 3(x-yi) = \frac{1-18i}{2-i} \times \frac{2+i}{2+i}$$

$$4x + 4yi - 3x + 3yi = \frac{1(2+i) - 18i(2+i)}{4 - i^2}$$

$$(4x - 3x) + (4yi + 3yi) = \frac{2+i - 36i - 18i^2}{4+1}$$

$$x + 7yi = \frac{2 - 35i + 18}{5}$$

$$x + 7yi = \frac{20 - 35i}{5}$$

$$\because i^2 = -1$$

$$x + 7yi = 4 - 7i$$

Compare real & imag

$$\boxed{x = 4}, \quad 7y = -7$$

$$y = \frac{-7}{7}$$

$$\boxed{y = -1}$$

$$\text{Hence } \boxed{Z = 4 - i}$$

Answer

UNIT #1 EXERCISE #1.1

$$x+yi \quad \overline{z} = x-yi$$

COMPLEX NUMBER

Q2 Find the Conjugate of Complex Numbers.

$$\because i^2 = -1$$

$$\sqrt{i^2} = \sqrt{-1}$$

$$i = \sqrt{-1}$$

i) $4-3i$

sol: $z = 4-3i$

$$\overline{z} = 4+3i$$

Ans

$$\overline{z} = 2 - \frac{1}{\sqrt{5}}i$$

Ans

ii) $3i+8$

$$z = 3i+8$$

$$\overline{z} = -3i+8$$

Ans

iii) $2 + \sqrt{\frac{-1}{5}}$

$$z = 2 + \frac{1}{\sqrt{5}}i$$

$$\overline{z} = 2 - \frac{1}{\sqrt{5}}i$$

Ans

iv) $\frac{5i}{2} - \frac{7}{8}$

$$z = \frac{5i}{2} - \frac{7}{8}$$

$$\overline{z} = -\frac{5}{2}i - \frac{7}{8}$$

Answer

UNIT #1 EXERCISE #1.1

7. Find the Magnitude of Complex

(i) $11 + 12i$

$$Z = 11 + 12i$$

$$|Z| = \sqrt{(11)^2 + (12)^2}$$

$$|Z| = \sqrt{121 + 144}$$

$$|Z| = \sqrt{265}$$

Answer

ii) $Z = (2 + 3i) - (2 + 6i)$

$$Z = 0 - 3i$$

$$|Z| = \sqrt{(0)^2 + (-3)^2}$$

$$= \sqrt{9}$$

$$|Z| = 3$$

$$\sqrt{8} \left(\begin{array}{r} 2 \overline{) 8} \\ \underline{2} \\ 0 \\ \underline{0} \\ 0 \end{array} \right)$$

$$\sqrt{2+2}$$

$$(2\sqrt{2})$$

iii) $(2 - i)(6 + 3i)$

$$Z = 2(6 + 3i) - i(6 + 3i)$$

$$Z = 12 + 6i - 6i - 3i^2$$

$$Z = 12 + 3$$

$$Z = 15 + 0i$$

$$|Z| = \sqrt{(15)^2}$$

$$|Z| = 15$$

iv) $\frac{3 - 2i}{2 + i}$

$$Z = \frac{3 - 2i}{2 + i} \times \frac{2 - i}{2 - i}$$

$$Z = \frac{3(2 - i) - 2i(2 - i)}{4 - i^2}$$

COMPLEX NUMBER

Numbers.

$$Z = \frac{6 - 3i - 4i + 2i^2}{4 - (-1)}$$

$$Z = \frac{6 - 7i - 2}{5}$$

$$Z = \frac{4}{5} - \frac{7i}{5}$$

$$|Z| = \sqrt{\left(\frac{4}{5}\right)^2 + \left(-\frac{7}{5}\right)^2}$$

$$= \sqrt{\frac{16}{25} + \frac{49}{25}}$$

$$|Z| = \sqrt{\frac{16 + 49}{25}} \Rightarrow |Z| = \frac{\sqrt{65}}{5}$$

v) $(\sqrt{3} - \sqrt{-8})(\sqrt{3} + \sqrt{-8})$

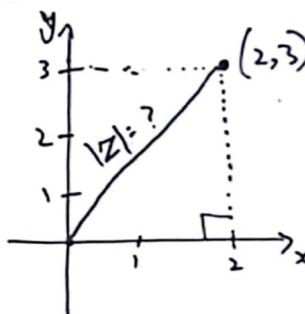
$$Z = (\sqrt{3} - 2\sqrt{2}i)(\sqrt{3} + 2\sqrt{2}i)$$

$$Z = (\sqrt{3})^2 - (2\sqrt{2}i)^2$$

$$Z = 3 - 8(-1)$$

$$Z = 3 + 8$$

$$Z = 11 + 0i$$



$$Z = x + yi$$

$$Z = 2 + 3i$$

$$|Z| = \sqrt{x^2 + y^2}$$

$$|Z| = \sqrt{x^2 + y^2}$$

$$|Z| = \sqrt{4 + 9} = \sqrt{13}$$

$$|Z| = \sqrt{(11)^2 + 0^2}$$

$$|Z| = 11$$

Answer